Stochastic Finance and Risk Modelling

Chapter 5: Multi-period model of financial market

Based on slides by Ioane Muni-Toke
Laurence Carassus and Gaoyue Guo

Expanded Study Resource

March 22, 2026

Contents

Introduction **2**

1 Discrete-Time Market Model **3**
1.1 The Multi-Period Model Setup . . . 3
1.1.1 Price Processes . . . 4
1.1.2 An Explicit Construction . . . 4
1.1.3 Discounted Processes . . . 5
1.2 Self-Financing Strategies and Portfolios . . . 5
1.2.1 Characterizations of Self-Financing Strategies . . . 6

2 Arbitrages and Martingale Measures **8**
2.1 Arbitrage . . . 8
2.2 Martingale Measures . . . 9
2.3 The Fundamental Theorem of Asset Pricing (FTAP) . . . 10

3 Payoffs and Derivatives **11**
3.1 Formalizing Payoffs . . . 11
3.1.1 Examples of Standard and Exotic Derivatives . . . 11
3.2 Replicable Payoffs . . . 12
3.3 The No-Arbitrage Price . . . 14

4 Complete Market **15**
4.1 Market Completeness . . . 15
4.2 The Second Fundamental Theorem of Asset Pricing . . . 15
4.2.1 Examples of Complete vs Incomplete Markets . . . 16

Introduction

This document serves as a comprehensive study guide for the course "Stochastic Finance and Risk Modelling". It expands upon the lecture slides relative to Chapter 5: "Multi-period model of financial market".

The goal of this text is to provide a rigorous mathematical foundation required for modern financial modeling in a multi-period discrete-time setting. We extend the concepts from one-period models to dynamic multi-period models. We cover the setup of the discrete-time market, the formal definition of self-financing strategies and portfolios, arbitrage opportunities, martingale measures, and the pricing and replication of European derivatives. The text concludes with an analysis of market completeness.

Chapter 1

Discrete-Time Market Model

In to financial modeling, transitioning from a single-period model to a multi-period model is essential for capturing the dynamic nature of trading. In this chapter, we lay out the mathematical foundations of a discrete-time multi-period market model.

1.1 The Multi-Period Model Setup

We consider a financial market consisting of one riskless asset (often called the money market account or a bank account) and $d \in \mathbb{N}$ risky assets (such as stocks or indices). Trading occurs at discrete time steps $t \in \{0, 1, \ldots, T\}$, where $T \in \mathbb{N}$ denotes the finite time horizon of the model.

To formalize the uncertainty inherent in financial markets, we introduce a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Ω is the sample space, representing the set of all possible market scenarios (or paths of asset prices) up to time T.
- $\mathcal{F}$ is the σ-algebra of events, representing all possible observable sets of scenarios.
- $\mathbb{P}$ is the historical or physical probability measure, representing the actual likelihood of market events occurring.

Crucially, in a dynamic multi-period model, information arrives progressively over time. This flow of information is mathematically modeled by a **filtration** $\mathbb{F} = (\mathcal{F}_t)_{t=0,\ldots,T}$. A filtration is an increasing sequence of sub-σ-algebras of $\mathcal{F}$, meaning that $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_T = \mathcal{F}$. Intuitively, $\mathcal{F}_t$ represents the set of all information available to investors up to and including time t. Any decision made at time t can only depend on information in $\mathcal{F}_t$; it cannot depend on the future.

Remark 1.1. At time $t = 0$, we assume that the initial prices of all assets, $(S_0^0, S_0^1, \ldots, S_0^d)$, are strictly observable and known. Because there is no uncertainty regarding the initial state of the market, the initial σ-algebra is trivial:

$$\mathcal{F}_0 = \{\emptyset, \Omega\}$$

This implies that any $\mathcal{F}_0$-measurable random variable is deterministically constant (almost surely).

1.1.1 Price Processes

Let S_t^i denote the price of the i-th asset at time t. The price evolution of asset i over time is represented by a stochastic process $S^i = (S_t^i)_{t=0,\dots,T}$.

For the model to be economically sensible, the price S_t^i must be known at time t. Mathematically, this means that the stochastic process S^i is **adapted** to the filtration $\mathbb{F}$. That is, for every $t \in \{0, \dots, T\}$, the random variable S_t^i is $\mathcal{F}_t$-measurable:

$$\sigma\left(S_t^1, \dots, S_t^d\right) \subseteq \mathcal{F}_t, \quad \forall t \in \{0, \dots, T\}.$$

1.1.2 An Explicit Construction

It is instructive to provide an explicit canonical construction for this abstract setup, focusing on a framework where the only source of randomness is the asset prices themselves.

The Riskless Asset: The riskless asset, denoted by S^0, models a bank account generating a constant, risk-free interest rate $r \geq 0$ per period. Assuming an initial price of $S_0^0 = 1$, the deterministic evolution of the riskless asset is defined as:

$$S_t^0 := (1+r)^t, \quad t = 0, \dots, T.$$

This represents discrete compounding where 1 unit of currency invested at $t = 0$ grows deterministically over time.

The Sample Space Ω: Since we have d risky assets traded over T periods, a complete path of the strictly positive risky asset prices can be represented as a vector in $\mathbb{R}_+^{d \times T}$. We can explicitly define the sample space Ω as the Cartesian product:

$$\Omega := \underbrace{\mathbb{R}_+^d \times \cdots \times \mathbb{R}_+^d}_{T \text{ times}}.$$

An element $\omega \in \Omega$ represents a specific price path: $\omega = (\omega_1, \dots, \omega_T)$, where each $\omega_t = (\omega_{t,1}, \dots, \omega_{t,d}) \in \mathbb{R}_+^d$ is the vector of prices for the d assets at time t. The "universe" $\mathcal{F}$ is given by the Borel σ-algebra $\mathcal{B}(\Omega)$.

Asset Price Functions: We then define the price of asset i at time t as the projection mapping from the entire path space Ω to the specific value at time t:

$$S_t^i : \Omega \to \mathbb{R}_+, \quad S_t^i(\omega) := \omega_{t,i}.$$

Writing $S_t = (S_t^1, \dots, S_t^d)$, we have $S_t(\omega) = \omega_t$.

The natural choice for the filtration in this explicit formulation is the natural filtration generated by the price processes themselves. We define:

$$\mathcal{F}_t := \sigma(S_1, \dots, S_t), \quad t = 1, \dots, T,$$

with $\mathcal{F}_T = \mathcal{F}$. Under this setup, $\mathcal{F}_t$ embodies "the history of prices up to time t," and no exogenous information is considered.

1.1.3 Discounted Processes

A common simplification in pricing theory is to eliminate the effect of the deterministic interest rate by expressing all prices relative to the riskless asset S^0. This is equivalent to choosing the bank account as the **numéraire**.

We define the **discounted price processes** $\tilde{S}^i$ for $i = 0, \ldots, d$:

$$\tilde{S}_t^i := \frac{S_t^i}{S_t^0} = \frac{S_t^i}{(1+r)^t}, \quad t = 0, \ldots, T.$$

In this discounted unit, the riskless asset's price is constant: $\tilde{S}_t^0 = 1$ for all t. By applying this transformation, we factor out the time value of money, dramatically simplifying the characterization of martingales and self-financing strategies.

1.2 Self-Financing Strategies and Portfolios

In a dynamic market, an investor continuously updates their portfolio by buying and selling assets according to a strategy. A trading strategy is formally defined by a $(d+1)$-dimensional stochastic process $\xi = (\xi^0, \xi^1, \ldots, \xi^d)$, where:

- ξ_t^0 represents the number of units held in the riskless asset between time $t-1$ and time t.
- ξ_t^i (for $i = 1, \ldots, d$) represents the number of units held in the i-th risky asset between time $t-1$ and time t.

A critical requirement is that ξ must be **predictable**. A process ξ is predictable relative to the filtration $\mathbb{F}$ if for every $t \in \{1, \ldots, T\}$, the random variable ξ_t is $\mathcal{F}_{t-1}$-measurable. Economic intuition dictates this: the portfolio allocation ξ_t chosen to be held during the interval $[t-1, t)$ must be decided at time $t-1$, using only the information available up to time $t-1$. It cannot foresee the prices that will be revealed at time t. For compactness, let us denote $\xi_t = (\xi_t^1, \ldots, \xi_t^d)$.

Definition 1.1 (Self-Financing Strategy)**.** A predictable trading strategy $(\xi^0, \xi) = (\xi_t^0, \xi_t)_{t=1,\ldots,T}$ taking values in $\mathbb{R}^{d+1}$ is called **self-financing** if it satisfies:

$$\left(\xi_t^0, \xi_t\right) \cdot \left(S_t^0, S_t\right) = \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(S_t^0, S_t\right) \quad \text{for all } t = 1, \ldots, T-1. \tag{1.1}$$

Equivalently:

$$\xi_t^0 S_t^0 + \sum_{i=1}^d \xi_t^i S_t^i = \xi_{t+1}^0 S_t^0 + \sum_{i=1}^d \xi_{t+1}^i S_t^i$$

Interpretation: Equation (1.1) perfectly captures the essence of a self-financing portfolio. At time t, before any rebalancing occurs, the investor holds the portfolio ξ_t which was decided at $t-1$. The value of this portfolio evaluated at the newly revealed prices S_t is the left-hand side: $(\xi_t^0, \xi_t) \cdot (S_t^0, S_t)$. After observing the prices, the investor decides to rebalance their portfolio to the new allocation ξ_{t+1}, which will be held from time t to $t+1$. The cost of assembling this new portfolio is exactly the right-hand side: $(\xi_{t+1}^0, \xi_{t+1}) \cdot (S_t^0, S_t)$.

The equality asserts that the value of the portfolio right before rebalancing exactly covers the cost of the newly assembled portfolio. There is no exogenous infusion of capital, nor is any capital withdrawn for consumption. All purchases are funded strictly by the sale of existing assets in the portfolio.

Definition 1.2 (Portfolio Value Process)**.** The portfolio associated with a self-financing strategy $(\xi_0, \xi) = (\xi_t^0, \xi_t)_{t=1,\dots,T}$ has an adapted value process $(V_t^{\xi_0,\xi})_{t=0,\dots,T}$ described by:

$$V_0^{\xi_0,\xi} = \left(\xi_1^0, \xi_1\right) \cdot \left(S_0^0, S_0\right) = \xi_1^0 + \sum_{i=1}^{d} \xi_1^i S_0^i,$$

$$V_t^{\xi_0,\xi} = \left(\xi_t^0, \xi_t\right) \cdot \left(S_t^0, S_t\right) = \xi_t^0 (1+r)^t + \sum_{i=1}^{d} \xi_t^i S_t^i, \quad t = 1, \dots, T.$$

Because the strategy is self-financing, any fluctuation in the portfolio's value is purely a consequence of changes in asset prices, not changes in position volumes. We can formally calculate the variation of the portfolio's value from time t to $t+1$:

$$\begin{aligned} V_{t+1}^{\xi_0,\xi} - V_t^{\xi_0,\xi} &= \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(S_{t+1}^0, S_{t+1}\right) - \left(\xi_t^0, \xi_t\right) \cdot \left(S_t^0, S_t\right) \\ &= \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(S_{t+1}^0, S_{t+1}\right) - \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(S_t^0, S_t\right) \quad \text{(using the self-financing relation)} \\ &= \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(\left(S_{t+1}^0, S_{t+1}\right) - \left(S_t^0, S_t\right)\right). \end{aligned}$$

This elegantly demonstrates that the change in portfolio value is completely attributed to the returns of the underlying assets (ΔS_{t+1}) multiplied by the predictable allocations (ξ_{t+1}).

1.2.1 Characterizations of Self-Financing Strategies

A crucial mathematical property is how self-financing strategies interact with discounted prices. Working with discounted prices often significantly eases derivations.

Proposition 1.2. *Let $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ be a trading strategy. The following claims are equivalent:*

(i) The strategy $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ is self-financing.

(ii) The self-financing condition holds when evaluated with discounted prices:

$$\left(\xi_t^0, \xi_t\right) \cdot \left(1, \tilde{S}_t\right) = \left(\xi_{t+1}^0, \xi_{t+1}\right) \cdot \left(1, \tilde{S}_t\right), \quad t = 1, \dots, T-1.$$

(iii) The discounted portfolio value $\tilde{V}_t^{\xi_0,\xi} := \frac{V_t^{\xi_0,\xi}}{S_t^0}$ can be expressed purely as stochastic discrete-time integral against the discounted risky asset price processes:

$$\tilde{V}_t^{\xi_0,\xi} = V_0^{\xi_0,\xi} + \sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}), \quad t = 0, \dots, T.$$

Proof. **(i) $\iff$ (ii):** Dividing the standard self-financing condition (1.1) by the strictly positive observable scalar S_t^0 immediately yields the condition in (ii), and multiplying by S_t^0 reverses it.

(i) $\iff$ (iii): Assume the strict self-financing criteria holds. We compute the variation in the discounted portfolio value at time t:

$$\begin{aligned}
\tilde{V}_t^{\xi_0,\xi} - \tilde{V}_{t-1}^{\xi_0,\xi} &= (\xi_t^0, \xi_t) \cdot \left(1, \tilde{S}_t\right) - (\xi_{t-1}^0, \xi_{t-1}) \cdot \left(1, \tilde{S}_{t-1}\right) \\
&= (\xi_t^0, \xi_t) \cdot \left(1, \tilde{S}_t\right) - (\xi_t^0, \xi_t) \cdot \left(1, \tilde{S}_{t-1}\right) \quad \text{(using property (ii) above)} \\
&= (\xi_t^0, \xi_t) \cdot \left(\left(1, \tilde{S}_t\right) - \left(1, \tilde{S}_{t-1}\right)\right) \\
&= \xi_t^0(1-1) + \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) \\
&= \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}).
\end{aligned}$$

Summing these differences consecutively from $k = 1$ to t transforms a telescoping sum into exactly the expression given in (iii):

$$\tilde{V}_t^{\xi_0,\xi} = \tilde{V}_0^{\xi_0,\xi} + \sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1})$$

Noting that $\tilde{V}_0 = V_0/S_0^0 = V_0/1 = V_0$, the result is proven.

The reverse implication proceeds mechanically. Setting $\tilde{V}_t$ by the sum dictates exactly that $\tilde{V}_t - \tilde{V}_{t-1} = \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1})$, which easily unwinds back into the definition of property (ii). □

The true significance of property (iii) is that the bank account position (ξ_t^0) strictly vanishes from the representation of the discounted portfolio's gains. This leads perfectly into the following corollary proposition:

Proposition 1.3. *Let $(\xi_t)_{t=1,\dots,T}$ be an $\mathbb{F}$-predictable process taking values in $\mathbb{R}^d$ representing the holdings of risky assets. For every initial capital $x \in \mathbb{R}$, there exists a* ***unique*** *predictable process $(\xi_t^0)_{t=1,\dots,T}$ taking values in $\mathbb{R}$ such that the composite process $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ forms a self-financing strategy starting with an initial portfolio value $V_0^{\xi_0,\xi} = x$.*

Because the risky asset weights completely dictate the aggregate portfolio accumulation when combined with the self-financing mandate, the money-market position acts simply as a residual balancing term. The initial capital x and the trades ξ_t uniquely define the value of the portfolio at all times:

$$V_t = x \cdot S_t^0 + S_t^0 \sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}).$$

To understand why x is multiplied by S_t^0, recall from Proposition 1.2 that the discounted portfolio value is bounded mathematically as $\tilde{V}_t = x + \sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1})$. Multiplying both sides by the numéraire S_t^0 to obtain the exact actual value V_t distributes the multiplication to x. Financially, the term $x \cdot S_t^0$ represents the initial capital capitalized to time t by the risk-free rate (as if it had been entirely and passively invested in the bank account), because we assume $S_0^0 = 1$. The second term represents the accumulated trading gains from the risky assets, correctly capitalized to current time t. Consequently, we often safely denote a uniquely defined portfolio path strictly by its initial capital and risky allocations, writing $V^{x,\xi}$.

Chapter 2

Arbitrages and Martingale Measures

A central premise of financial mathematics and asset pricing is that properly functioning liquid markets do not afford arbitrage opportunities. In this chapter, we formalize the concept of an arbitrage and introduce the most critical theoretical tool in modern finance: the equivalent martingale measure. We connect these two ideas through the Fundamental Theorem of Asset Pricing.

2.1 Arbitrage

Broadly speaking, an arbitrage is a "free lunch": an investment strategy that requires zero initial capital, has absolutely no risk of losing money, and yields a strictly positive probability of making a profit.

Definition 2.1 (Arbitrage Opportunity)**.** A self-financing strategy (ξ_0, ξ) yields an **arbitrage opportunity** if the corresponding portfolio's value process $V^{\xi_0,\xi}$ satisfies the following three conditions:

1. $V_0^{\xi_0,\xi} = 0$: The strategy requires zero initial wealth.
2. $\mathbb{P}[V_T^{\xi_0,\xi} \geq 0] = 1$: There is zero probability of ending up with a negative portfolio value at the final time T (almost sure non-negativity).
3. $\mathbb{P}[V_T^{\xi_0,\xi} > 0] > 0$: There is a strictly positive probability of ending up with a strictly positive portfolio value.

A market model in which no such strategies exist is said to satisfy the **No-Arbitrage (NA) condition**. Historically, if an arbitrage exists, market participants would aggressively exploit it, driving prices to a new equilibrium where the arbitrage disappears. Therefore, to model markets accurately, one strictly requires the NA condition:

$$\text{No Arbitrage Condition:} \quad V_T^{0,\xi} \geq 0 \ \mathbb{P}\text{-a.s. for some } \xi \implies V_T^{0,\xi} = 0 \ \mathbb{P}\text{-a.s.}$$

While defined over the entire interval $[0, T]$, arbitrage can be isolated to single-period segments. The next proposition shows that if an arbitrage exists over the multi-period horizon, one must be able to construct a one-step arbitrage.

Proposition 2.1. *There exists an arbitrage opportunity in the market if and only if there exists some specific trading period $t \in \{1, \dots, T\}$ and an $\mathcal{F}_{t-1}$-measurable random vector ξ_t taking values in $\mathbb{R}^d$ such that:*

$$\begin{cases} \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) \geq 0 & \mathbb{P}\text{-}a.s., \\ \mathbb{P}\left[\xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) > 0\right] > 0. \end{cases} \tag{2.1}$$

Proof Sketch. The "if" direction is straightforward. Suppose at time t we can construct the one-period arbitrage ξ_t. We define a full strategy where from $k = 0$ to $t - 1$, we hold nothing ($\xi_k = 0$). At t, we hold the arbitrage-generating portfolio ξ_t and perfectly offset its cost with the risk-free asset. The generated gain is precisely $\tilde{V}_t = \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1})$. Since it requires zero initial wealth and strictly generates positive expected gain with no downside risk, we then hold this cumulative gain in the riskless asset until time T ($\xi_k = 0$ for $k > t$). The final portfolio value V_T will be $(1 + r)^{T-t} V_t$, which trivially satisfies the global arbitrage conditions.

The "only if" direction shows that if a global T-period arbitrage exists, you can analyze the portfolio's discounted increments frame by frame backwards in time. For the aggregated gain $\sum_{k=1}^{T} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1})$ to be strictly positive with no downside, there must be at least one period where an increment is unilaterally strictly advantageous, meaning the NA condition must strictly hold at every individual trading period. □

2.2 Martingale Measures

The profound insight of stochastic finance is that the absence of arbitrage is deeply connected to the existence of a specific probability measure underpinning the market.

Definition 2.2 (Martingale Measure / Risk-Neutral Measure)**.** A probability measure $\mathbb{Q}$ defined on the measurable space $(\Omega, \mathcal{F})$ is said to be a **martingale measure** (or a **risk-neutral measure**) if the discounted price processes $(\tilde{S}_t^1, \dots, \tilde{S}_t^d)_{t=0,\dots,T}$ are martingales under $\mathbb{Q}$. Mathematically, this means:

1. Integrability: $\mathbb{E}_{\mathbb{Q}}[\tilde{S}_t^i] < \infty$ for all t and i.

2. Martingale property: Under the filtration $\mathbb{F}$, we have

$$\mathbb{E}_{\mathbb{Q}}\left[\tilde{S}_t^i \mid \mathcal{F}_{t-1}\right] = \tilde{S}_{t-1}^i \quad \text{for } 1 \leq t \leq T.$$

This mathematical property bears an intuitive economic meaning. Under a martingale measure, the expected future discounted price of any asset, given all current information $\mathcal{F}_{t-1}$, is exactly structurally equal to its current discounted price. Therefore, under $\mathbb{Q}$, no asset offers an expected excess return (risk premium) relative to the riskless bond. The measure $\mathbb{Q}$ "neutralizes" all market risk preferences.

For this measure to be relevant to the actual market governed by the physical probability $\mathbb{P}$, $\mathbb{Q}$ and $\mathbb{P}$ must agree on what is "possible" and what is "impossible".

Definition 2.3 (Equivalent Martingale Measure)**.** A martingale measure $\mathbb{Q}$ is called an **equivalent martingale measure (EMM)** if it is equivalent to the physical measure $\mathbb{P}$, denoted as $\mathbb{Q} \sim \mathbb{P}$. Equivalency means the two measures share exactly the same null sets:

$$\text{For any } A \in \mathcal{F}, \quad \mathbb{P}[A] = 0 \iff \mathbb{Q}[A] = 0.$$

This equivalence is vital. It guarantees that an event is almost surely impossible (e.g., losing money on an arbitrage) under the real-world measure $\mathbb{P}$ if and only if it is almost surely impossible under the pricing measure $\mathbb{Q}$.

2.3 The Fundamental Theorem of Asset Pricing (FTAP)

The link between absence of arbitrage and martingale measures constitutes one of the most celebrated theorems in financial mathematics. Historically established by Harrison and Pliska (1981), and later expanded into its general continuous-time form by Delbaen and Schachermayer (1994), it provides the backbone of modern pricing theory.

Theorem 2.2 (First Fundamental Theorem of Asset Pricing)**.** *The market is arbitrage-free (i.e., there are no arbitrage opportunities) if and only if there exists an equivalent martingale measure* $\mathbb{Q}$*.*

Intuition and Implications. The robust mathematical proof involves advanced functional analysis (specifically, separating hyperplane theorems like the Hahn-Banach theorem applied to the convex cones of replicable claims), but its direct financial implications are easy to see.

Assume an EMM $\mathbb{Q}$ exists. Take any self-financing strategy starting with $V_0 = 0$. By Proposition 1.2, the discounted portfolio value is:

$$\tilde{V}_T = \sum_{k=1}^{T} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}).$$

Since ξ_k is predictable and $\tilde{S}_k$ is a martingale under $\mathbb{Q}$, the integral of a predictable process against a martingale remains a martingale. Thus, $(\tilde{V}_t)$ is a $\mathbb{Q}$-martingale. Therefore, the expected value under $\mathbb{Q}$ of the final portfolio is equal to its initial value:

$$\mathbb{E}_{\mathbb{Q}}[\tilde{V}_T] = \tilde{V}_0 = 0.$$

If this strategy were an arbitrage, we would have $\mathbb{P}[V_T \geq 0] = 1$ and $\mathbb{P}[V_T > 0] > 0$. By equivalence ($\mathbb{P} \sim \mathbb{Q}$), this heavily implies $\mathbb{Q}[V_t \geq 0] = 1$ and $\mathbb{Q}[V_T > 0] > 0$. However, a non-negative random variable that is strictly positive with some probability must have a strictly positive expectation: $\mathbb{E}_{\mathbb{Q}}[\tilde{V}_T] > 0$. This severely contradicts $\mathbb{E}_{\mathbb{Q}}[\tilde{V}_T] = 0$. Hence, we conclude that no such arbitrage can possibly exist if an EMM $\mathbb{Q}$ exists. $\square$

Chapter 3

Payoffs and Derivatives

With the foundational model of a multi-period arbitrage-free market established, we can now turn to the pricing and hedging of financial contracts. These contracts are generically called derivatives because their value is "derived" from the underlying assets $S^1, \ldots, S^d$.

3.1 Formalizing Payoffs

Definition 3.1 (Payoff and European Derivative)**.** A **payoff** is a random payout delivered at a specific future time. Mathematically, it is defined as an $\mathcal{F}_T$-measurable random variable H. A **European derivative** written on the underlying assets $S^1, \ldots, S^d$ is a financial contract that guarantees its holder a payoff H at the maturity date T, where $H = F(S^0, S^1, \ldots, S^d)$ for some measurable function F.

The measurability condition $H \in \mathcal{F}_T$ simply ensures that by the time the contract expires at T, its value is perfectly known and unambiguously determined by the market history up to that point.

Broadly, European options can be classified based on their functional dependence on the price history:

- **European Vanilla Derivatives:** The payoff depends explicitly only on the prices at the final maturity date T: $H = F(S_T^0, S_T^1, \ldots, S_T^d)$. The path taken to reach S_T is irrelevant to the payout.
- **Exotic Derivatives (Path-Dependent):** The payoff H depends on the entire trajectory of the asset prices $(S_t)_{t=0,\ldots,T}$. While still settled exclusively at T, determining the payout requires looking back at the entire path.

3.1.1 Examples of Standard and Exotic Derivatives

To ground these definitions, we list several classical derivatives. For simplicity, assume they are written on a single underlying asset S^i. Let $K > 0$ be a fixed constant known as the strike price.

European Vanilla Options:

- **European Call Option:** Gives the right (but not the obligation) to buy the asset at price K at time T. The rational holder will exercise precisely when $S_T^i > K$.

$$H = (S_T^i - K)_+ = \max(S_T^i - K, 0)$$

- **European Put Option:** Gives the right to sell the asset at price K at time T. The holder exercises if the asset crashes below the strike.

$$H = (K - S_T^i)_+ = \max(K - S_T^i, 0)$$

- **Basket Call Option:** Written on a linearly weighted basket of all d risky assets with fixed weights w_i.

$$H = \left(\sum_{i=1}^{d} w_i S_T^i - K \right)_+$$

Path-Dependent (Exotic) Options:

- **Asian Call Option:** The payoff depends on the arithmetic average price of the asset over the contract's life. This smoothens out volatility and makes the option generally cheaper than a vanilla call, while protecting the holder from potential market manipulation strictly at maturity.

$$H = \left(\frac{1}{T} \sum_{t=1}^{T} S_t^i - K \right)_+$$

- **Lookback Call Option:** The payoff depends on the optimal (maximum) price reached by the asset during the contract's life. By definition, this will always pay out at least as much as a vanilla call, making it strictly more expensive.

$$H = \left(\max_{t=1,\dots,T} S_t^i - K \right)_+$$

- **Barrier (Up-and-Out) Call Option:** The payoff is identical to a vanilla call option, provided the asset's price never touches or breaches an upper barrier B during the entire life of the contract. If at any time $S_t^i \geq B$, the option is immediately knocked out and becomes worthless ($H = 0$).

$$H = (S_T^i - K)_+ \mathbf{1}_{\{S_t^i < B, \forall t=1,\dots,T\}}$$

Notice how indicator functions $\mathbf{1}_{\{\cdot\}}$ naturally model structural contingencies in path-dependent exotic options.

3.2 Replicable Payoffs

The central paradigm of modern derivative pricing relies on the concept of **replication**. If a derivative's random payoff can be perfectly emulated by actively trading the underlying market assets dynamically, then the derivative provides no inherently new risk to the market.

Definition 3.2 (Replicable Payoff)**.** A European payoff $H \in \mathcal{F}_T$ is said to be **replicable** (or **attainable**) if there exists a self-financing strategy $(\xi^0, \xi) = (\xi_t^0, \xi_t)_{t=1,\dots,T}$ such that the terminal value of its portfolio exactly matches the derivative's payoff in every possible scenario:

$$H = V_T^{\xi_0,\xi} = \left(\xi_T^0, \xi_T\right) \cdot \left(S_T^0, S_T\right) \quad \mathbb{P}\text{-a.s.}$$

This specific strategy is called the **replicating strategy** (or hedging strategy) for H.

If H is replicable, determining its fair price is conceptually straightforward using the No Arbitrage principle. If two assets (or portfolios) yield identical cash flows at all future states, they must have the exact same price today; otherwise, an investor could short the more expensive one, buy the cheaper one, and pocket a riskless initial difference. Therefore, the fair price of the payoff H at time $t = 0$ must precisely equal the initial capital required to set up its replicating portfolio, $V_0^{\xi_0,\xi}$.

Proposition 3.1. *Assume the market satisfies the No-Arbitrage condition (hence at least one equivalent martingale measure $\mathbb{Q}$ exists). Let H be a replicable European payoff. Then, for* ***any*** *replicating strategy (ξ_0, ξ) and for* ***any*** *equivalent martingale measure $\mathbb{Q}$, the discounted value of the replicating portfolio satisfies:*

$$\tilde{V}_t^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0} \,\middle|\, \mathcal{F}_t\right] \quad \mathbb{Q}\text{-}a.s. \text{ for } t = 0, \dots, T. \tag{3.1}$$

In particular, the discounted portfolio process $(\tilde{V}_t^{\xi_0,\xi})_{t=0,\dots,T}$ is a true $\mathbb{F}$-adapted martingale under $\mathbb{Q}$.

Proof and Derivation. Let $\mathbb{Q}$ be an equivalent martingale measure. By Proposition 1.2, the discounted value process for any self-financing strategy is given by the stochastic integral:

$$\tilde{V}_T^{\xi_0,\xi} = \tilde{V}_t^{\xi_0,\xi} + \sum_{k=t+1}^{T} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}).$$

Taking the conditional expectation under $\mathbb{Q}$ given $\mathcal{F}_t$:

$$\mathbb{E}^{\mathbb{Q}}\left[\tilde{V}_T^{\xi_0,\xi} \mid \mathcal{F}_t\right] = \tilde{V}_t^{\xi_0,\xi} + \sum_{k=t+1}^{T} \mathbb{E}^{\mathbb{Q}}\left[\xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}) \mid \mathcal{F}_t\right].$$

To understand this critical step, we must evaluate the expectation $\mathbb{E}^{\mathbb{Q}}\left[\xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}) \mid \mathcal{F}_t\right]$ for a future time $k > t$. Because the trading strategy ξ_k is decided at time $k-1$ (it is $\mathcal{F}_{k-1}$-measurable or predictable), it is not yet known at our current time t. Therefore, we cannot simply factor ξ_k out of the expectation conditioned on $\mathcal{F}_t$. Instead, we apply the **tower property of conditional expectation** to first condition on the intermediate time $k-1$, taking advantage of the fact that $\mathcal{F}_t \subseteq \mathcal{F}_{k-1}$:

$$\mathbb{E}^{\mathbb{Q}}\left[\xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}) \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}\left[\xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}) \mid \mathcal{F}_{k-1}\right] \,\middle|\, \mathcal{F}_t\right]$$

In the inner expectation conditioned on $\mathcal{F}_{k-1}$, the variable ξ_k behaves as a known constant and can be factored out. Furthermore, because the discounted price $\tilde{S}$ is a $\mathbb{Q}$-martingale, the expected price change from $k-1$ to k is perfectly zero: $\mathbb{E}^{\mathbb{Q}}\left[\tilde{S}_k - \tilde{S}_{k-1} \mid \mathcal{F}_{k-1}\right] = 0$. The expected increments therefore evaluate progressively to zero:

$$\cdots = \mathbb{E}^{\mathbb{Q}}\left[\xi_k \cdot \underbrace{\mathbb{E}^{\mathbb{Q}}\left[(\tilde{S}_k - \tilde{S}_{k-1}) \mid \mathcal{F}_{k-1}\right]}_{=0} \,\middle|\, \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}[\xi_k \cdot 0 \mid \mathcal{F}_t] = 0$$

Hence, $\mathbb{E}^{\mathbb{Q}}\left[\tilde{V}_T^{\xi_0,\xi} \mid \mathcal{F}_t\right] = \tilde{V}_t^{\xi_0,\xi}$. This demonstrates that $\tilde{V}$ is a martingale.

Because the strategy strictly replicates the payoff, $V_T^{\xi_0,\xi} = H$ almost surely. Dividing by the numéraire gives $\tilde{V}_T^{\xi_0,\xi} = H/S_T^0$. Substituting this into our martingale expression yields the final result:

$$\tilde{V}_t^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0} \,\middle|\, \mathcal{F}_t\right].$$

□

Evaluating this at $t = 0$, where $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is trivial, the conditional expectation collapses to a simple expectation. The initial price of the replicating portfolio is therefore fixed and deterministic:

$$V_0^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right].$$

3.3 The No-Arbitrage Price

In the broader context—what if a derivative is **not** replicable? Can we still price it without introducing inconsistencies into the market?

Definition 3.3 (No-Arbitrage Price)**.** Assume the market modeling the primary assets $(S^0, \ldots, S^d)$ is arbitrage-free. We formally call $\pi_0 \in \mathbb{R}$ an **arbitrage-free price** of an European payoff H if one can artificially augment the original market with a newly traded asset X defined by:

$$X_0 = \pi_0 \quad \text{and} \quad X_T = H$$

such that the structurally expanded market consisting of $d+2$ assets $(S^0, S^1, \ldots, S^d, X)$ remains strictly arbitrage-free.

By invoking the First Fundamental Theorem of Asset Pricing on this augmented market, standard integration yields a comprehensive characterization of all possible prices that do not violate No Arbitrage.

Proposition 3.2. *The set of all valid arbitrage-free prices for any bounded European payoff H takes the exact functional form:*

$$\left\{\mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right] : \mathbb{Q} \text{ is an equivalent martingale measure for } S, \text{ such that } \mathbb{E}^{\mathbb{Q}}\left[\left|\frac{H}{S_T^0}\right|\right] < \infty\right\}.$$

This set represents an interval of prices $[p_{min}, p_{max}]$. If you sell the derivative for more than p_{max}, you introduce an arbitrage in the market (you priced it too high); if you sell it for less than p_{min}, you also create an arbitrage (you priced it too low).

Critically, when a payoff H **is** replicable, its initial capital V_0^{ξ} is completely unaffected by the subjective choice of the measure $\mathbb{Q}$! Every EMM $\mathbb{Q}$ must agree on $\mathbb{E}^{\mathbb{Q}}[H/S_T^0]$.

Proposition 3.3. *Assume the NA condition holds. Let H be a strongly replicable European derivative. Then H has a **unique** arbitrage-free price:*

$$V_0^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right]$$

*where (ξ_0, ξ) is **any** replicating portfolio and $\mathbb{Q}$ is **any** equivalent martingale measure. The set of valid prices naturally collapses to a single unique point.*

Chapter 4

Complete Market

In Chapter 3, we saw that when a payoff is replicable, we can objectively determine its exact unique and fair arbitrage-free price. A natural and exceedingly important question arises: are *all* payoffs replicable?

4.1 Market Completeness

Different market models have vastly different expressive capabilities when it comes to hedging risks. Some models perfectly allow you to offset any possible arbitrary financial exposure strictly by trading the underlying assets dynamically.

Definition 4.1 (Complete Market)**.** A market is formally called **complete** if **every** possible bounded European payoff $H \in \mathcal{F}_T$ is replicable by some self-financing strategy.

If a market is complete and arbitrage-free, derivatives become entirely redundant from an economic perspective. The market is "complete" in the sense that introducing a new derivative contract does not expand the universe of obtainable random payoffs—investors could have implicitly synthesized that exact derivative themselves using dynamic trading. In these ideal complete markets, every derivative can be perfectly priced and perfectly hedged, meaning an investment bank can theoretically eliminate absolutely all market risk from issuing options.

4.2 The Second Fundamental Theorem of Asset Pricing

Market completeness is not just an abstract property; much like the First Fundamental Theorem linked absence of arbitrage to the mere *existence* of a martingale measure, the Second Fundamental Theorem powerfully links completeness to the *uniqueness* of that measure.

Theorem 4.1 (Second Fundamental Theorem of Asset Pricing)**.** *Assume a market model satisfies the No-Arbitrage condition (so at least one equivalent martingale measure exists). Then the market model is* ***complete*** *if and only if the equivalent martingale measure* $\mathbb{Q}$ *is* ***strictly unique****.*

Intuition and Interpretation. We have established earlier (Proposition 3.3) that if a payoff H is replicable, then its initial price must unambiguously be exactly $V_0^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[H/S_T^0\right]$ for

every equivalent martingale measure $\mathbb{Q}$. This means that all valid measures $\mathbb{Q}_1, \mathbb{Q}_2, \ldots$ must necessarily agree on the expectation of H.

If the market is arbitrarily complete, then *every* payoff is replicable. Thus, for any two equivalent martingale measures $\mathbb{Q}_1$ and $\mathbb{Q}_2$, they must identically yield:

$$\mathbb{E}^{\mathbb{Q}_1}\left[\frac{H}{S_T^0}\right] = \mathbb{E}^{\mathbb{Q}_2}\left[\frac{H}{S_T^0}\right]$$

for **all** bounded $\mathcal{F}_T$-measurable variables H. By standard measure theory, if two probability measures give the identical expectation for every bounded measurable function, then they are mathematically the same measure: $\mathbb{Q}_1 \equiv \mathbb{Q}_2$. Hence, $\mathbb{Q}$ is undeniably unique.

Conversely, if the market specifically admits multiple distinct equivalent martingale measures $\mathbb{Q}_1 \neq \mathbb{Q}_2$, then those bounds on arbitrage-free prices for unreplicable options will strictly not collapse to a single point. There will naturally be some bounded payoff H such that $\mathbb{E}^{\mathbb{Q}_1}[H] \neq \mathbb{E}^{\mathbb{Q}_2}[H]$. Since replicating portfolios must always command uniform expectations, this payoff cannot possibly be strongly replicable, and the market explicitly becomes **incomplete**. □

4.2.1 Examples of Complete vs Incomplete Markets

To put this theoretical result into perspective:

- **The Cox-Ross-Rubinstein (CRR) Binomial Model:** If an asset price can only move deterministically to two possible states (Up or Down) at each discretetime step, the market is completely spanned by trading a stock and a bond. The linear system has exactly two equations (one for Up, one for Down) and two unknowns (ξ_t^0, ξ_t^1). It perfectly resolves, predicting a unique EMM. Thus, the classic Binomial Model is **complete**.

- **The Trinomial Model:** If the asset price can strictly jump to three distinct states (Up, Middle, Down), but you still only have two assets (stock and bond) to hedge with, your system is heavily underdetermined. Three equations cannot be solved reliably with only two variables. In this model, you cannot uniquely replicate the price, and an infinite gamut of structurally valid EMMs will naturally arise. The Trinomial Model is intrinsically **incomplete**.

- **Stochastic Volatility Models:** Expanding into continuous time, models where variance fluctuates unpredictably (like the Heston model) introduce new distinct sources of random noise unspanned by the underlying asset. Unless the market also physically trades variance swaps or similar contracts, these models universally admit multiple EMMs and are incomplete.

In real-world empirical finance, markets are fundamentally incomplete. Frictional forces like transaction costs, unpredictable liquidity drops, short-selling restrictions, and continuous unhedgeable jump risks violently break the theoretical assumptions. Nonetheless, evaluating complete baseline models yields the critical foundations for pricing and hedging most standard instruments.